

## **CLASS & OBJECT**

1. A school admissions team needs to store and print each student's ID, name, department, and marks using separate records that behave like objects.
2. A billing counter must calculate the area of different rectangular floor mats from their length and breadth entered at runtime.
3. A workshop tool needs to compute the area and circumference of circular plates from user-provided radii.
4. An HR portal stores employee ID, name, designation, and salary, then prints a formatted employee directory.
5. A car showroom keeps an inventory of cars with brand, model, year, and price and displays all cars currently available.
6. A mobile store records brand, storage, RAM, and price for each device and prints a catalog.
7. A banking app maintains account number, customer name, and balance while supporting deposit and withdrawal actions.
8. A library catalog stores book ID, title, author, and category and displays all books added during the session.

## **INHERITANCE**

1. A zoo system has common animal behavior such as eat and sleep, while dogs, cats, and birds add species-specific actions.
2. A vehicle management app shares registration number and fuel type across all vehicles, while cars add seating capacity and trucks add load capacity.
3. A college database stores common person details and extends them for students with roll numbers and marks.
4. A company payroll module stores employee details and extends managers with a bonus component and supervisors with an allowance component.
5. A drawing tool defines common shape properties and lets circles and rectangles calculate their own areas.
6. A banking system defines generic account operations while savings accounts add interest calculation and current accounts add overdraft handling.

7. A school information system models a person and extends it into teacher and student roles with different behaviors.
8. A product catalog stores generic product details and extends electronics with warranty information and clothing with size information.

#### **POLYMORPHISM**

1. A calculator screen should accept two-number addition for simple users and three-number addition for advanced users without changing the action name.
2. A geometry app should call the same area() operation on circles, rectangles, and triangles and let each shape compute its own result.
3. A zoo simulation should call sound() on different animals and get species-specific sounds.
4. A vehicle simulator should call start() on bikes, cars, and trucks and let each type behave differently.
5. A payment gateway should invoke pay() on UPI, credit card, and wallet objects and process payments through the appropriate mechanism.
6. A banking app should call withdraw() on savings and current accounts and enforce account-specific rules.
7. A graphics engine should call draw() on different shapes and let each render itself differently.

#### **ENCAPSULATION**

1. A student portal must prevent marks from being modified directly and allow updates only through validation methods.
2. A banking application must hide the balance field and expose only deposit, withdraw, and balance-check operations.
3. An employee payroll system must keep salary private and permit changes only through authorized methods.
4. A mobile inventory app must prevent direct edits to product price and enforce updates through pricing rules.
5. A car showroom must expose price information through accessor methods while preventing arbitrary direct modification.

6. A hospital system must keep patient records private and reveal only approved information through service methods.

#### **ABSTRACTION**

1. A zoo app defines a generic animal concept and requires every animal type to provide its own sound behavior.
2. A graphics system defines a generic shape contract and requires each concrete shape to implement area calculation.
3. A vehicle control system defines common operations such as start and stop but lets bikes and cars implement them differently.
4. A company payroll system defines salary calculation abstractly and lets full-time and contract employees compute salary differently.
5. A payment framework defines a payment contract and allows UPI, credit card, and wallet implementations.
6. A smart-home platform defines a device interface and lets lights, fans, and AC units implement their own behavior.
7. A notification service should call send() on email, SMS, and push-notification objects and dispatch through different channels.